NAVRACHANA UNIVERSITY

Teaching Plan

Artificial Intelligence (CSE607)

Credit : 3

Faculty	Name	Contact	Office Hours	Office Location
	Jaideepsinh Raulji	jaideep.raulji@nuv.ac.in		

School	SET

Semester	Spring (Jan to June 2026)			
Lecture time & Weedays	SECTION_1	Monday	12:00:00-13:00:00	05-01-2026 TO 08-05-2026
		Tuesday	12:00:00-13:00:00	05-01-2026 TO 08-05-2026
		Thursday	11:00:00-12:00:00	05-01-2026 TO 08-05-2026
	SECTION_2	Monday	11:00:00-12:00:00	05-01-2026 TO 08-05-2026
		Friday	15:00:00-16:00:00	05-01-2026 TO 08-05-2026
		Thursday	14:00:00-15:00:00	05-01-2026 TO 08-05-2026

Pre-requisite	

Co-requisite	

Anti-requisite	

Course Description	This course studies how to realize the intelligent human behaviours on a computer. The ultimate goal is to make a computer that can learn, plan, and solve problems autonomously. The course offers a comprehensive curriculum that delves into the cutting-edge fields of Artificial Intelligence. Students will explore techniques such graph search techniques, probablistic reasoning, learning algorithms, neural networks, convolutional neural networks, recurrent neural networks, generative adversarial networks, Markov Decision Processes, dynamic programming, Monte Carlo methods, and temporal difference learning. Additionally, the program will cover natural language processing topics, computer vision, and generative models. The program also emphasizes ethical considerations in AI, including bias, fairness, privacy, and societal impact. By combining theoretical knowledge with practical applications, students will gain hands-on experience through research projects, preparing them for a successful career in the field of artificial intelligence.

Course Abstract	-

Course Objectives	1. To provide an understanding of entire computer System, mathamatical foundations, software agents, environment. 2. To give brief knowledge about searching methodologies. 3. To provide depth knowledge about probabilistic reasoning and reinforcement learning algorithms. 4. To aware students about indepth mathmatical intuition about machine and neural network algorithms. 5. To make student familiar about recent technologies in the field of Data, Natural Language Processing and Computer Vision.

Course Outcomes	CO1. Gain understanding of complex computing systems, comparing human computer intelligence systems, mathematical foundations, software agents and environment. CO2. Gaining knowledge about informed and uninformed and adversarial searching methodologies. CO3. Gaining depth knowledge about probablistic reasoning and reinforcement learning algorithms. CO4. Students will learn about indepth mathematical intuition about machine and neural learning algorithms. CO5. Student will gain familiarity of recent developments in AI technologies like Data, NLP and CV.

Course Typology	Theory

Course Outline	**Unit-1 Introduction:** Introduction, History, Foundations and Mathematical treatments, Problem solving with AI, AI models, Learning aspects in AI, What is an intelligent Agents, Rational agent, Environments types, types of Agents, Problem Formulations, Review of tree and graph structures, State space representation, Search graph and Search tree Introduction to AI (No change) Introduction, History, Foundations and Mathematical treatments, Problem solving with AI, AI models, learning aspects in AI, what is an intelligent Agents, Rational agent, Environment types, types of Agents, Problem Formulations, Review of tree and graph structures, State space representation, Search graph and Search tree **Unit-2 Search Algorithms:** Random search, Search with closed and open list, Depth first and Breadth first search, Heuristic search, Best first search, A* algorithm, Game Search Search Algorithms (No Change) Random search, Search with closed and open list, Depth first and Breadth first search, Heuristic search, best first search, A* algorithm, Minimax Search (Game search) **Unit-3 Probabilistic Reasoning:** Probability, conditional probability, Bayes Rule, Bayesian Networks- representation, construction and inference, temporal model, hidden Markov model. Probabilistic Reasoning: Probability, conditional probability, Bayes Rule, Bayesian Networks- representation, construction and inference, temporal model, hidden Markov model. Reinforcement Learning: Q-Learning, Markov Decision Processes (MDPs), dynamic programming algorithm, Monte Carlo methods, deep reinforcement learning. **Unit-4 Learning Algorithms**: Supervised and Unsupervised learning techniques, Linear and Non-linear machine learning algorithms, dimensionality reduction techniques, model selection and hyperparameter tuning, Artificial Neural Network, ANN, Forward and Backward propagation, Multilayer Perceptron Model (MLP), Deep neural networks, convolutional neural networks, recurrent neural networks, generative adversarial networks, transformer models, transfer learning and fine-tuning **Unit-5 Communicating, Perceiving, and Acting:** Learning Algorithms, Language Models, Text Classification, Information Retrieval, Information Extraction, Syntactic Analysis (Parsing), Machine Translation, Speech Recognition, Image Formation and object recognition, Early Image-Processing Operations, Object Recognition by Appearance Natural Language Processing: Text preprocessing and feature extraction, language models, sentiment analysis, text classification, machine translation, text summarization Computer Vision: Image processing techniques, object detection and recognition, image and video analysis, generative models for image and video synthesis Artificial Intelligence Ethics: Ethical considerations in AI development and deployment, bias and fairness in AI, privacy and security concerns, societal impact of AI

Pedagogy	Theory, Discussions and Experiential

Expectations from Students	Pre-reads from previous sessions

Assessment / Evaluation	

Project / Assignment Details	Project submission in groups (3 to 4 students) at the end of semester.
	1. Write a program in Python to find factorial of a number using a loop. Also find the same using a recursive function. Implement this creating both the function in a class.
	2. Write a python program to implement stack and queue using OOP paradigm.
	3. Write a python program to create a binary tree, add elements, retrieve elements using pre-order, post-order and in-order traversal.
	4. Write a Program in Python to solve tic-tac-toe problem implementing minimax algorithm.
	5. Write a program in Python for Breadth First Search.
	6. Write a pythoa program to implement Breadth First Search and Depth First Search algorithm on following graph. Consider start node as A.
	7. Write a program in Python to implement Depth First Search.
	8. Implement BFS and DFS on following graph.
	9. Write a program in Python for Best First Search
	10. Write a python program to implement Uniform Cost Search with cumulative cost.
	11. Write a program in Python for A* Search
	12. Write a program to implement Depth Limited Search and Iterative Deepening Search on following graph/tree.
	graph = {
	'0': ['4', '8'],
	'1': ['3', '5'],
	'8': ['9'],
	'3': ['10'],
	'5': ['11'],
	'9': ['12'],
	'10': [],
	'11': [],
	'12': []
	}
	13. Write a program to implement UCS on following graph.
	graph = {
	'A': {'B': 1, 'C': 2},
	'B': {'D': 3, 'E': 4},
	'C': {'F': 5},
	'D': {'G': 6},
	'E': {'G': 7},
	'F': {'G': 8},
	'G': {}
	}
	start = 'A'
	goal = 'G'
	14. Write a program to implement Best First Search on following graph.
	graph = {
	'A': ['B',12, 'C',14}, #heuristic value A to H is 13, B to H is 12, C to H is 4
	'B':{'D':11, 'E':10},
	'C':{'F':6, 'G':7},
	'D':{'H':0},
	'E':{'H':0},
	'F':{'H':0},
	'G':{'H':0}
	}
	15. Write a program to implement A* search on following graph.
	graph = {
	'A': {'B': [3,8], 'C': [2,9]},
	'B': {'D': [3,7], 'E': [4,6]},
	'C': {'F': [5,4]},
	'D': {'G': [6,0]},
	'E': {'G': [9,0]},
	'F': {'G': [6,0]},
	'G': {}
	}
	start = 'A'
	goal = 'G'
	16. Write a program to implement A* search on the following graph.
	Note : Structure of data : "Key Name " : {"City Name": Latitude, Longitude, Heuristic value}
	graph = {
	'START': {'Jammu': [32.7266,74.8570,1600)},
	'Jammu': {'Amritsar': [31.6339, 74.8722,1400], 'Delhi': [28.7040,77.1024,1300]},
	'Amritsar': {'Sri-Gangaganar': [29.9094,73.8800,1340], 'Jodhpur': [26.2389,73.0243, 1230]},
	'Delhi': {'Jaipur': [26.9124, 75.7873,1000],'Gwalior'-[26.2124, 78.1772,1100]},
	'Sri-Gangaganar': {'Udaipur': [24.5854, 73.7125,400]},
	'Jodhpur': {'Himmatnagar': [23.5969, 72.9630,300]},
	'Jaipur': {'Kota': [25.2138, 75.8648,300]},
	'Gwalior': {'Ratlam': [23.3315,75.0367,250]},
	'Udaipur': {'Vadodara': [22.3072,73.1812,0]},
	'Himmatnagar': {'Vadodara': [22.3072,73.1812,0]},
	'Kota': {'Vadodara': [22.3072,73.1812,0]},
	'Ratlam': {'Vadodara': [22.3072,73.1812,0]},
	}
	start = 'Jammu'
	goal = 'Vadodara'
	HINT:
	#pip install geopy
	import geopy.distance
	coords_1 = (22.3071, 73.1812) #Vadodara
	coords_2 = (23.0225, 72.5713) #Ahmedabad
	print ("distance in kms is ", geopy.distance.geodesic(coords_1, coords_2).km)
	17. Implement the following.
	18. Write a program in Python for calculating conditional probability for following data in CSV file. The input columns are light blue coloured, remaining are calculative.
	year Students with Job at Campus P(Job) Students who learnt python P(Py) Students with python and job P(Job^py) Con-P(Job|Py) Con-P(Py|Job)
	2015 28 0.56 15 0.3 10 0.2 0.666666667 0.357142857
	2016 32 0.64 21 0.42 17 0.34 0.80952381 0.53125
	2017 34 0.68 25 0.5 21 0.42 0.84 0.617647059
	2018 37 0.74 34 0.68 31 0.62 0.911764706 0.837837838
	2019 38 0.76 39 0.78 37 0.74 0.948717949 0.973684211
	2020 46 0.92 44 0.88 42 0.84 0.954545455 0.913043478
	19. Naive Bayes classification from scratch using Excel for below given tabular data.
	20. Naive Bayes classification using python sklearns lib for below given tabular data.
	Sr. No. Color Type Origin Stolen
	1 Red SUV Domestic Yes
	2 Red SUV Imported Yes
	3 Red Sports Imported Yes
	4 Red Sports Domestic No
	5 Red Sports Imported Yes
	6 Yellow SUV Imported Yes
	7 Yellow SUV Imported Yes
	8 Yellow SUV Domestic No
	9 Red SUV Imported Yes
	10 Red Sports Imported No
	11 Yellow Sports Imported Yes/No ??
	21. Create a model to predict next word conditional probability-based prediction model for Gujarati language (Download gujarati text from sources available on internet)
	22. Create a model to predict whether a person will have car or not based on dataset attached using Naive Bayes Classifier. (user_data_cars_1.csv)
	1. Calculate Entropy and Gini for following dataset in Excel. (playplaynot.csv)
	2. Write a python script to implement Decision Tree classifier on same dataset. (playplaynot.csv)
	3. Write python script to implement Random Forest classifier on following dataset. (iris.csv)
	Attachment playplaynot.csv, iris.csv, ML Observation Table.docx
	23. Write a python script to implement
	1. KNN Classifier and
	2. KNN Regression
	based on match on 3 attached datasets.
	Record your observations with different parameters in the ML record sheet
	attached.
	Upload code and ML Observation table.
	Data set attached : user_data_cars_1.csv, pima-indiana-diabetes.csv, cars.csv
	24. Write a python script to implement
	1. Regression using KNN, Linear, Ridge, Lasso and ElasticNet on cars.csv dataset to predict CO2 emission.
	2. Classification using LogisticRegression on pima-indiana-diabetes.csv.
	25. Develop a ML model to predict Quality of Milk (Low, Medium, High) from the given dataset (Milk_Quality.csv).
	Perform following operations
	1. Read the dataset.
	2. Display the shape of dataset
	3. Display columns of dataset.
	4. Check for null values.
	5. Show descriptive statistics of dataset.
	6. Display unique values in each column (for pH, Temp, etc)
	7. Draw hist plots for each column.
	8. Remove outliers if required.
	9. Balance the dataset equally for the target output variable by removing or augmenting records.
	10. Using K-Best or any Feature selection technique, use the best X features.
	11. Perform scaling or encoding on features.
	12. Create multiple models.
	13. Select the most appropriate model to host on web creating a web-api and consume.
	26. Develop a ML model to predict car price from the given dataset (usedcars.csv).
	Perform following operations
	1. Read the dataset.
	2. Display the shape of dataset
	3. Display columns of dataset.
	4. Check for null values.
	5. Show descriptive statistics of dataset.
	6. Display unique values in each column.
	7. Draw hist plots for each column.
	8. Remove outliers if required.
	9. Using K-Best or any Feature selection technique, use the best X features.
	10. Perform scaling or encoding on features.
	11. Create multiple models.
	12. Select the most appropriate model to host on web creating a web-api and consume.
	27. Write a python script to transliterate between hindi and Gujarati and vice-versa.
	Please find unicode chart
	https://www.ssec.wisc.edu/~tomw/java/unicode.html
	28. Write a Python script for language transliteration between Gujarati and English Script.
	Input : ???? ??? ??????? ?????? ???? ???? ????? ??.
	output : Aapde badha krutrim buddhi vishay sikhi rahya chee.
	29. Write an Object-Oriented Program which reads texts from a file. It must display file statistics a below.
	a. No. of sentences.
	b. No. of words.
	c. No. of total characters (Does not include whitespace)
	d. No. of whitespaces
	e. Total no. of digits, uppercase and lowercase letters.
	30. Write an Object Oriented Program which creates vocabulary of words and also counts each word in a document.
	Eg. Content
	The birds are flying. The boy is walking. The Ganges are great river system. The Narmada river flows from rift valley.
	output :
	[(The,3), (birds,1), (are,1), (birds,1), (are,2), (flying,1), (boy,1), (river,2)]
	31. Develop an NLP application which tokenizes text, removes punctuation marks, converts to lower case, removes spelling errors, removes stopwords, convert to root word using either stemmer or lemmatizer and displays counts/frequency of the main text words.
	32. Write a program for next word prediction using N-Gram conditional probability.
	33. Write an script to build Bag-of-Word and TF-IDF model from English text.
	34. Write a program to create a POS tagger for English/Hindi/Gujarati language using DL model.
	35. Write a NLP application which classifies Gujarati news to categories (Entertainment, tech, business).
	36. Write an NLP application which classifies message as Spam or Not-Spam (Ham).
	37. Write a program to create word embeddings for English / Local language.
	38. Write an AI-NLP application to classify whether an message is spam or not. Develop ANN-MLP model with different no. of layers, no. of nodes in a layer, optimizer function, loss function and activation function. Finally mention the best model and hyperparameters for your problem.
	Mention below parameters in a tabular format
	a) No. of input neurons.
	b) No. of layers
	c) No. of nodes in each hidden layer
	d) Activation function used in hidden layer and output layer neurons.
	e) Optimization function used in hidden layer and output layer neurons.
	f) loss function used in hidden layer and output layer neurons.
	g) No. of Epochs
	h) Accuracy with each parameters.
	39. Write a python program to perform edge detection on image.
	40. A program to perform face detection on image.
	41. To implement reinforcement learning using Q-Learning.

Course Material	Reference Books: 1. Stuart Russes, Peter Norvig, Artificial Intelligence: A Modern Approach by Stuart Russell and Peter Norvig, Pearson Education 2. Christopher Bishop, Pattern Recognition and Machine Learning, Springer Publications. 3. Ian Goodfellow, Yoshua Bengio, et al, Deep, MIT Press 4. Kevin Murphy, Machine Learning , MIT Press 5. Daphne Koller, Nir Friedman, Probabilistic Graphical Models: Principles and Techniques MIT Press 6. Michael Nielsen, Neural Networks and Deep Learning, Nielsen Press 7. Daniel Jurafsky, James H. Martin, Speech and Language Processing, Pearson Education 8. Richard Szeliski, Computer Vision: Algorithms and Applications, Springer publications
Additional Information	-

Session Plan

Topic Title	Session No.	Topic & Subtopic Details	Readings, Cases, etc.	Activities	Important Dates
Introduction	1	Introduction, History, Foundations and Mathematical treatments, Problem solving with AI	Stuart Russell and Peter Norvig, "Artificial Intelligence: A Modern Approach", Ch.1,2	Lecturing and Class Discussions	-
"	2	AI models, Learning aspects in AI	Stuart Russell and Peter Norvig, "Artificial Intelligence: A Modern Approach", Ch.1,2	"	-
"	3	What is an intelligent Agents, Rational agent, Environments types, types of Agents	Stuart Russell and Peter Norvig, "Artificial Intelligence: A Modern Approach", Ch.1,2	"	-
"	4	Problem Formulations, Review of tree and graph structures	Stuart Russell and Peter Norvig, "Artificial Intelligence: A Modern Approach", Ch.3, 4, 5	"	-
"	5	State space representation, Search graph and Search tree	Stuart Russell and Peter Norvig, "Artificial Intelligence: A Modern Approach", Ch. 3, 4, 5	"	-
"	6	State space representation, Search graph and Search tree	Stuart Russell and Peter Norvig, "Artificial Intelligence: A Modern Approach", Ch. 3, 4, 5	Lecturing, Discussions and Application Demo	-
Search Algorithms	7	Random search	Stuart Russell and Peter Norvig, "Artificial Intelligence: A Modern Approach", Ch. 3, 4, 5	"	-
"	8	Search with closed and open list	Stuart Russell and Peter Norvig, "Artificial Intelligence: A Modern Approach", Ch. 3, 4, 5	"	-
"	9	Search with closed and open list	Stuart Russell and Peter Norvig, "Artificial Intelligence: A Modern Approach", Ch. 3, 4, 5	"	-
"	10	Depth first and Breadth first search	Stuart Russell and Peter Norvig, "Artificial Intelligence: A Modern Approach", Ch. 3, 4, 5	"	-
"	11	Heuristic search, Best first search	Stuart Russell and Peter Norvig, "Artificial Intelligence: A Modern Approach", Ch. 3, 4, 5	"	-
"	12	A* algorithm, Game Search	Stuart Russell and Peter Norvig, "Artificial Intelligence: A Modern Approach", Ch. 3, 4, 5	"	-
"	13	A* algorithm, Game Search	Stuart Russell and Peter Norvig, "Artificial Intelligence: A Modern Approach", Ch. 3, 4, 5	"	-
Probabilistic Reasoning	14	Probability, conditional probability	Stuart Russell and Peter Norvig, "Artificial Intelligence: A Modern Approach", Ch. 14-17	"	-
"	15	Probability, conditional probability	Stuart Russell and Peter Norvig, "Artificial Intelligence: A Modern Approach", Ch. 14-17	"	-
	16	Bayes Rule, Bayesian Networks- representation	Stuart Russell and Peter Norvig, "Artificial Intelligence: A Modern Approach", Ch. 14-17	"	-
"	17	Probability based algorithms	Stuart Russell and Peter Norvig, "Artificial Intelligence: A Modern Approach", Ch. 14-17	"	-
"	18	hidden Markov model	Stuart Russell and Peter Norvig, "Artificial Intelligence: A Modern Approach", Ch. 14-17	"	-
"	19	Learning Algorithms - Linear and Non-linear models	Stuart Russell and Peter Norvig, "Artificial Intelligence: A Modern Approach", Ch. 14-17	"	-
"	20	Learning Algorithms - Linear and Non-linear models	Stuart Russell and Peter Norvig, "Artificial Intelligence: A Modern Approach", Ch. 14-17	"	-
"	21	Learning Algorithms - Linear and Non-linear models	Stuart Russell and Peter Norvig, "Artificial Intelligence: A Modern Approach", Ch. 14-17	"	-
"	22	Learning Algorithms - Linear and Non-linear models	Stuart Russell and Peter Norvig, "Artificial Intelligence: A Modern Approach", Ch. 14-17	"	-
"	23	Learning Algorithms - Linear and Non-linear models	Stuart Russell and Peter Norvig, "Artificial Intelligence: A Modern Approach", Ch. 14-17	"	-
Communicating, Perceiving, and Acting	24	Text Processing models	Stuart Russell and Peter Norvig, "Artificial Intelligence: A Modern Approach", Ch.22-24	"	-
"	25	Text Processing models	Stuart Russell and Peter Norvig, "Artificial Intelligence: A Modern Approach", Ch. 22-24	"	-
"	26	Text Processing models, RNN, LSTM, Tranformers	Stuart Russell and Peter Norvig, "Artificial Intelligence: A Modern Approach", Ch. 22-24	"	-
"	27	Information Retrieval, Information Extraction, Syntactic Analysis (Parsing)	Stuart Russell and Peter Norvig, "Artificial Intelligence: A Modern Approach", Ch. 22-24	"	-
"	28	Information Retrieval, Information Extraction, Syntactic Analysis (Parsing)	Stuart Russell and Peter Norvig, "Artificial Intelligence: A Modern Approach", Ch. 22-24	"	-
"	29	Machine Translation, Speech Recognition	Stuart Russell and Peter Norvig, "Artificial Intelligence: A Modern Approach", Ch. 22-24	"	-
"	30	Machine Translation, Speech Recognition	Stuart Russell and Peter Norvig, "Artificial Intelligence: A Modern Approach", Ch. 22-24	"	-
"	31	Image Formation and object recognition	Stuart Russell and Peter Norvig, "Artificial Intelligence: A Modern Approach", Ch. 22-24	"	-
"	32	Image Formation and object recognition	Stuart Russell and Peter Norvig, "Artificial Intelligence: A Modern Approach", Ch. 22-24	"	-
"	33	Early Image-Processing Operations	Stuart Russell and Peter Norvig, "Artificial Intelligence: A Modern Approach", Ch. 22-24	"	-
"	34	Modelling with CNN	Stuart Russell and Peter Norvig, "Artificial Intelligence: A Modern Approach", Ch 22-24	"	-
	35	Using pre-trained vision models			
	36	Using pre-trained vision models			
Markov Decision process & Reinforcement Learning	37	Introduction to Reinforcement Learning	Stuart Russell and Peter Norvig, "Artificial Intelligence: A Modern Approach", Ch. 14-17	"	-
"	38	temporal difference learning, active reinforcement learning-Q learning	Stuart Russell and Peter Norvig, "Artificial Intelligence: A Modern Approach", Ch. 14-17	"	-
"	39	temporal difference learning, active reinforcement learning-Q learning	Stuart Russell and Peter Norvig, "Artificial Intelligence: A Modern Approach", Ch. 18-21	"	-
"	40	policy iteration and partially observable MDPs	Stuart Russell and Peter Norvig, "Artificial Intelligence: A Modern Approach", Ch. 18-21	"	-
"	41	Passive reinforcement learning, direct utility estimation	Stuart Russell and Peter Norvig, "Artificial Intelligence: A Modern Approach", Ch. 18-21	"	-
"	42	adaptive dynamic programming	Stuart Russell and Peter Norvig, "Artificial Intelligence: A Modern Approach", Ch. 18-21	"	-
"	43	Revision	Stuart Russell and Peter Norvig, "Artificial Intelligence: A Modern Approach"	"	-
"	44	Revision	Stuart Russell and Peter Norvig, "Artificial Intelligence: A Modern Approach"	"	-
"	45	Revision	Stuart Russell and Peter Norvig, "Artificial Intelligence: A Modern Approach"	"	-

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